

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical

Practical – II

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AIM: Describing the Distribution of a Variable:

A. Obtain a dataset and calculate descriptive statistics (mean, median, mode, variance, etc.) for a specific variable of interest.

CODE:

```
import pandas as pd
print("Library Loaded.")
```

OUTPUT:

Library Loaded.

CODE:

```
df = pd.read_csv("Walmart.csv")
print("Dataset Loaded.")
```

OUTPUT:

... Dataset Loaded.

CODE:

```
df.head()
```

OUTPUT:

...	Store	Date	Weekly_Sales	Holiday_Flag	Temperature	Fuel_Price	CPI	Unemployment
0	1	05-02-2010	1643690.90	0	42.31	2.572	211.096358	8.106
1	1	12-02-2010	1641957.44	1	38.51	2.548	211.242170	8.106
2	1	19-02-2010	1611968.17	0	39.93	2.514	211.289143	8.106
3	1	26-02-2010	1409727.59	0	46.63	2.561	211.319643	8.106
4	1	05-03-2010	1554806.68	0	46.50	2.625	211.350143	8.106

CODE:

```
sales = df["Weekly_Sales"]
print("Weekly sales stored in sales.")
```

OUTPUT:

Weekly sales stored in sales.

CODE:

```

print("Mean :", sales.mean())
print("Median :", sales.median())
print("Minimum :", sales.min())
print("Maximum :", sales.max())
print("Range :", sales.max() - sales.min())
print("Variance :", sales.var())
print("Count :", sales.count())
print("Sum :", sales.sum())
print("\nDescriptive Statistics")
print(sales.describe())
print("\nMode :")
print(sales.mode())

```

OUTPUT:

```

... Mean : 1046964.8775617715
Median : 960746.04
Minimum : 209986.25
Maximum : 3818686.45
Range : 3608700.2
Variance : 318509684088.30096
Count : 6435
Sum : 6737218987.11

```

Descriptive Statistics

```

count  6.435000e+03
mean    1.046965e+06
std     5.643666e+05
min     2.099862e+05
25%     5.533501e+05
50%     9.607460e+05
75%     1.420159e+06
max     3.818686e+06
Name: Weekly_Sales, dtype: float64

```

Mode :

```

0      209986.25
1      213538.32
2      215359.21
3      219804.85
...
6432   3749057.69
6433   3766687.43
6434   3818686.45

```

```
Name: Weekly_Sales, Length: 6435, dtype: float64
```

B. Create visualizations (histograms, box plots) to depict the distribution of a variable and analyze its characteristics.

CODE:

```
#Practical 2B
import matplotlib.pyplot as plt
import seaborn as sns
print("Libraries Loaded.")
```

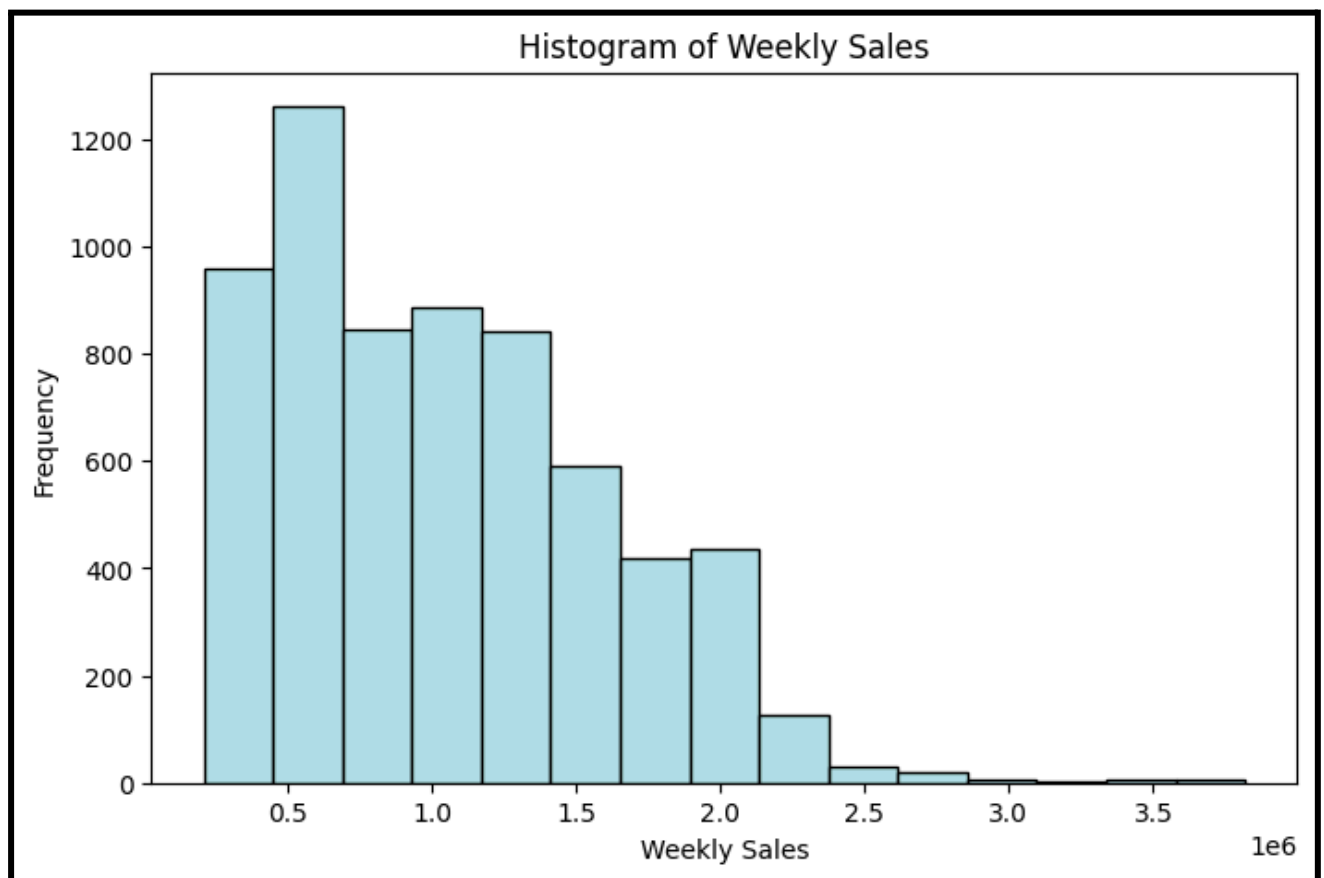
OUTPUT:

Libraries Loaded.

CODE:

```
plt.figure(figsize=(8,5))
plt.hist(sales, bins=15, edgecolor="black" , color = "PowderBlue")
plt.title("Histogram of Weekly Sales")
plt.xlabel("Weekly Sales")
plt.ylabel("Frequency")
plt.show()
```

OUTPUT:



CODE:

```
plt.figure(figsize=(8,3))

sns.boxplot(x=sales, color="Yellow")

plt.title("Box Plot of Weekly Sales")
plt.xlabel("Weekly Sales")

plt.show()
```

OUTPUT:

